

Lead Scoring Project - X Education

Student: Gowtham Kumar

Objective: Predict lead conversion and recommend calling strategies.

Dataset & Preprocessing

Original shape: (9240, 37)

Dropped columns: Prospect ID, Lead Number

Modeling Approach

Preprocessing: Scaling numerical, One-Hot for categoricals

Model: Logistic Regression (L1)

Feature selection: Coefficients from L1 model

Key Metrics

Accuracy: 0.937
Precision: 0.931
Recall: 0.904
ROC-AUC: 0.978

Top Features

1. Tags_Lost to EINS (coef=9.7511)
2. Tags_Closed by Horizzon (coef=7.6634)
3. Tags_Will revert after reading the email (coef=5.5807)
4. Tags_Unknown (coef=3.6761)
5. Lead Quality_Worst (coef=-2.5709)
6. Tags_switched off (coef=-2.4504)

Business Recommendations

Aggressive phase: Lower threshold, tiered calling lists, interns on Tier B.

Conservative phase: Raise threshold, call only high-probability leads, use email for others.